A Comprehensive Survey of NLP Tools for Ge'ez-Script Languages: Systematic Analysis of 16 Tools and 70 Papers (2016–2026)

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Abstract

This comprehensive systematic review analyzes 70 peer-reviewed papers (2016–2026) using PRISMA methodology to discover, catalog, and analyze 19 foundational NLP tools for Ge'ez-script languages (Amharic, Tigrinya, Classical Ge'ez). Organized across four technical pillars—tokenization, morphology, embeddings/language models, and sequence labeling—our analysis reveals: (a) improving but persistent underrepresentation of Classical Ge'ez with 3 tools (16% coverage); (b) severe bottleneck in sequence labeling with just 1 tool (5% of all tools); (c) decreasing but dominant single-language approach with 63% of tools targeting individual languages; (d) balanced preference with 58% rule-based and 42% neural methods. We trace tool development acceleration from 1 tool/year (2016–2020) to 3.0 tools/year (2021–2023), with P3 (embeddings) showing strongest growth (4→6 tools). Critical gaps remain: sequence labeling severely underexplored, and cross-lingual transfer learning limited (37% multilingual). Based on quantitative findings and qualitative analysis of 70 papers, we propose a five-priority research agenda for coordinated, script-aware NLP development: (1) unified tokenization framework, (2) cross-lingual morphological transfer, (3) script-optimized embeddings, (4) expanded sequence labeling, and (5) community infrastructure. This survey serves as a foundation for future research and practical reference for developers, researchers, and policymakers advancing Ge'ez-script language technology.

Keywords: Ge'ez script, systematic review, low-resource NLP, Amharic, Tigrinya, Classical Ge'ez, foundational tools, PRISMA methodology, language technology

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1 Introduction

1.1 Motivation and Context

Natural language processing (NLP) has experienced exponential growth over the past decade, with major advances in tokenization, morphological analysis, embeddings, and sequence labeling. However, these advances have accrued unevenly across world languages. Ge'ez-script languages—comprising Amharic (primary language of Ethiopia, 32M speakers), Tigrinya (primary language of Eritrea, 7M speakers), and Classical Ge'ez (liturgical language, 0.5M active users)—represent a critical gap in global NLP coverage. With over 62 million speakers combined, Ge'ez-script languages warrant systematic, coordinated development of foundational NLP tools.

Despite their linguistic complexity and digital significance, Ge'ez-script languages remain critically underrepresented in major NLP conferences, funding mechanisms, and development frameworks. The Ge'ez script itself presents unique challenges: (1) syllabic structure (unlike Latin alphabets), (2) complex morphology with root-pattern systems, (3) limited digital resources and corpora, and (4) diverse use cases from religious texts to modern social media.

1.2 Prior Surveys and Positioning

Previous surveys have addressed multilingual NLP (Joshi et al., 2020) and low-resource language technology (Blodgett et al., 2021), but none specifically focuses on tool-centered systematic analysis for Ge'ez-script languages. This survey fills this gap by:

1. Conducting the first tool-centered systematic review specific to Ge'ez-script NLP
2. Adopting a script-level linguistic perspective rather than isolated language perspective
3. Quantifying disparities in language coverage, technical focus, and development patterns
4. Proposing a concrete, prioritized research agenda based on systematic evidence

1.3 Survey Structure and Scope

This paper follows PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) methodology. Our scope includes:

- **Languages:** Amharic, Tigrinya, Classical Ge'ez (and combinations)
- **Temporal:** 2016–2026 (10-year window reflecting modern NLP era)
- **Peer review:** Only peer-reviewed conference/journal papers (ACL, EMNLP, NAACL, LREC, COLING, etc.)
- **Tools:** Publicly available, reproducible implementations (datasets, models, frameworks)
- **Exclusions:** Preprints, non-peer-reviewed work, downstream applications without foundational contribution

2 Methodology

2.1 Literature Search and Screening

We conducted systematic searches across seven major venues: ACL, EMNLP, NAACL, EACL, AACL, COLING, and LREC. Search terms included: "Amharic NLP", "Tigrinya NLP", "Ge'ez language", "Ethiopic script", "Ge'ez-script languages", "low-resource African languages", and combinations.

2.1.1 Inclusion Criteria

- Peer-reviewed publication (conference or journal)
- Contribution to Ge'ez-script language processing
- Foundational impact (tokenization, morphology, embeddings, sequence labeling, or datasets/resources)
- English-language publication or translation available

2.1.2 Exclusion Criteria

- Preprints or non-peer-reviewed work
- Pure downstream applications (e.g., sentiment analysis models) without methodological novelty
- Duplicate or substantially similar papers
- Non-English languages without translation

2.2 Tool Discovery Strategy

In parallel to literature review, we conducted tool discovery across five major repositories:

1. **GitHub:** Primary open-source repository (web scraping, keyword search)
2. **Zenodo:** Academic research datasets and models
3. **PyPI:** Python package ecosystem
4. **HuggingFace Hub:** Pre-trained models and datasets
5. **DBLP:** Linked datasets from academic papers

For each tool, we extracted 30 metadata fields: - Basic info: name, description, version, release date, DOI - Technical: programming language, dependencies, computational requirements - Linguistic: supported languages, script encoding, morphological features - Capabilities: specific functions (tokenization, POS tagging, morphology, embeddings, etc.) - Coverage: domains, data sizes, performance metrics - Accessibility: license, documentation quality, community support

2.3 Deduplication and Verification

We performed semantic deduplication using: 1. Exact name matching 2. URL similarity (Levenshtein distance) 3. Semantic similarity (0.75 threshold on TF-IDF vectors)

Each tool was manually verified for: - Ge'ez-script language relevance (scope verification) - Functionality and reproducibility - Active maintenance status - Documentation completeness

2.4 Data Extraction and Analysis

From 70 papers, we extracted: - Bibliographic: authors, title, year, venue, DOI - Technical: algorithms, models, frameworks - Linguistic: languages, script handling, morphological features - Contributions: datasets, tools, benchmarks - Performance: metrics, baselines, comparative results

For 16 tools, we extracted: - Classification: type (dataset, model, tool, resource) - Technical pillar: P1 (tokenization), P2 (morphology), P3 (embeddings), P4 (sequence labeling), or Other - Language coverage: which Ge'ez-script languages supported - Design patterns: monolingual vs. multilingual, rule-based vs. neural - Development timeline: release dates and update frequency

3 Results

3.1 Literature Review Findings

Our comprehensive search yielded 70 peer-reviewed papers analyzing Ge'ez-script NLP across multiple topics:

- Multilingual NLP (15 papers): General frameworks applicable to Ge'ez
- Low-resource language technology (12 papers): Methodology for under-resourced languages
- Morphological analysis (8 papers): Ge'ez-specific morphology and root systems
- Machine translation (10 papers): Especially English-Amharic, English-Tigrinya
- Embeddings and language models (12 papers): Word2vec, FastText, BERT variants
- Neural NLP architectures (8 papers): Sequence-to-sequence, attention mechanisms
- Treebanks and corpora (5 papers): Universal Dependencies, linguistic annotation
- Other applications (10 papers): Sentiment analysis, POS tagging, named entity recognition

3.2 Tool Inventory: 19 Ge'ez-script Tools

3.2.1 Overview Statistics

Across all repositories, we identified 19 foundational tools with the following characteristics:

- **Language coverage:** Amharic 9 (47%), Tigrinya 11 (58%), Classical Ge'ez 3 (16%)
- **Technical pillars:** P1 (4, 21%), P2 (3, 16%), P3 (6, 32%), P4 (1, 5%), Other (5, 26%)
- **Tool types:** Dataset (7, 37%), Model (6, 32%), Tool (4, 21%), Resource (2, 10%)
- **Source repositories:** GitHub (11, 58%), Zenodo (5, 26%), HuggingFace (2, 11%), PyPI (1, 5%)
- **Design patterns:** Single-language (12, 63%), Multilingual (7, 37%), Rule-based (11, 58%), Neural (8, 42%)
- **Development timeline:** 2016–2020 (5 tools, 1.0/year), 2021–2023 (9 tools, 3.0/year), 2024–2026 (5 tools, acceleration)

3.2.2 Language Representation Analysis

Figure 1: Language Distribution of Ge'ez-script Tools
Tool development reveals persistent underrepresentation of Classical Ge'ez, though growth is occurring across all languages.

Language	Tools	%	Coverage Status
Amharic	9	47%	Well-researched
Tigrinya	11	58%	Well-researched
Classical Ge'ez	3	16%	**Improving but still underrepresented**
Total	**19**	**100%**	

Analysis: Classical Ge'ez has grown from 2 to 3 tools (12% to 16%), showing improvement. However, it remains underrepresented relative to its liturgical and cultural importance. Overlaps exist (some tools support multiple languages).

Figure 1: Language distribution of 19 Ge'ez-script NLP tools (2016–2026). Growth across all languages, with Classical Ge'ez improving but still underrepresented.

3.2.3 Technical Pillar Analysis

3.2.4 Design Pattern Analysis

3.2.5 Development Timeline Analysis

3.3 Complete Tool Inventory Table

4 Detailed Analysis

4.1 Pillar 1: Tokenization

Tokenization—the first essential step in NLP pipelines—breaks raw text into meaningful units (words, subwords, morphemes). For Ge'ez-script languages, tokenization is partic-

Figure 2: Technical Pillar Distribution
P3 (embeddings) now leads with 32%, but P4 (sequence labeling) remains critical bottleneck at only 5%.

Pillar	Tools	%	Status	
P1: Tokenization	4	21%	Foundation (established, growing)	
P2: Morphology	3	16%	Foundation (stable)	
P3: Embeddings/LMs	6	32%	Strong growth momentum	
P4: Seq. Labeling	1	5%	**SEVERE BOTTLENECK**	
Other: Infrastructure	5	26%	Data & resources	
Total	**19**	**100%**		

Analysis: P3 (embeddings/LMs) has grown substantially (4→6 tools, 25

Figure 2: Technical pillar distribution reveals severe P4 (sequence labeling) bottleneck limiting system development.

ularly complex due to:

1. **Syllabic structure:** Ge'ez script is syllabic (not purely alphabetic), requiring script-aware processing
2. **Word boundaries:** Modern Amharic/Tigrinya uses spaces inconsistently; classical texts use different conventions
3. **Punctuation:** Multiple punctuation systems (Ethiopian comma , colon , question mark)
4. **Compound words:** Complex morphology creates ambiguity in word boundaries

Current tools (4): Amharic-Tokenizer, Morph LSTM, Text Normalization, and MoVoC provide foundational coverage. MoVoC's morphology-aware approach represents progress, but comprehensive script-aware frameworks handling all Ge'ez-script languages remain needed. Future work should develop unified tokenization handling syllabic script properties and encoding variations.

4.2 Pillar 2: Morphology

Ge'ez morphology is exceptionally complex, featuring:

1. **Root-pattern system:** Semitic root + multiple patterns generate word families (e.g., Amharic = S-D system)
2. **Agglutinative features:** Affixes modify semantic and grammatical roles

Figure 3: Design Patterns in Tool Development
63% single-language (improving toward multilingual), 58% rule-based (neural approaches gaining).

Dimension	Count	%	Strategy	
Single-language	12	63%	Depth-first (language-specific)	
Multilingual	7	37%	Emerging cross-lingual transfer	
Rule-based	11	58%	Interpretability & explainability	
Neural	8	42%	Growing data-driven approaches	

Analysis: Shift toward multilingual tools (31%→37%) and neural approaches (35%→42%) indicates community maturation. Rule-based methods (58%) still dominate, reflecting strong linguistic expertise and interpretability preferences in Ge'ez-script NLP.

Figure 3: Design patterns reveal predominantly single-language and rule-based approaches in Ge'ez-script NLP tools.

3. **Gemination:** Consonant doubling changes meaning (Amharic vs.)
4. **Person/number/gender marking:** Rich verbal morphology

Current tools (3): HornMorpho, Semitic Law, Translation provide rule-based analysis but lack neural approaches. Only 3 tools for morphology represents critical gap for a language family with such complex morphology.

4.3 Pillar 3: Embeddings and Language Models

Embeddings capture semantic and syntactic information, essential for downstream NLP. Current progress:

1. **Monolingual embeddings:** Word2vec and FastText for Amharic; limited Tigrinya coverage
2. **Pre-trained language models:** Tigrinya-PLMs represents growing momentum; BERT variants emerging
3. **Multilingual approaches:** Limited; mBERT/XLM-R underperform on Ge'ez scripts compared to Latin alphabets

Six tools (Tigrinya-PLMs, TLMD, Amharic WSD, Amharic Corpus, LGSE Embeddings, Word Embedding Transfer) indicate significant growth in embeddings infrastructure. LGSE Embeddings and Word Embedding Transfer represent important advances in cross-lingual approaches. However, Classical Ge'ez coverage remains weak (only in LGSE), and cross-lingual transfer optimization remains largely unexplored.

Figure 4: Tool Development Timeline (2016–2026)
Three phases with accelerating growth: Foundation (1/year), Acceleration (3/year), Recent momentum.

Period	Years	Tools	Rate & Notes	
Early (Foundation)	2016–2020	5	∼1.0/year (community building)	
Middle (Acceleration)	2021–2023	9	∼3.0/year (momentum growth)	
Recent (Momentum)	2024–2026	5	Continued acceleration	
Total	**2016–2026**	**19**	Cumulative growth	

Analysis: Tool development shows strong acceleration from foundation phase (1/year) to acceleration phase (3/year), indicating growing international interest, funding, and community capacity. P3 (embeddings) drove recent growth.

Figure 4: Cumulative tool development timeline (2016–2026) shows acceleration from foundation phase to recent momentum.

4.4 Pillar 4: Sequence Labeling

Sequence labeling—POS tagging, named entity recognition (NER), chunking—enables structured understanding of text. Critical bottleneck:

1. **Only 1 tool** currently addresses sequence labeling for Ge'ez-script languages (Abusive Language classification)
2. **No POS taggers** publicly available for Tigrinya or Classical Ge'ez
3. **Limited NER** despite importance for information extraction and knowledge base construction
4. **Dependency parsing:** Minimal coverage; Universal Dependencies treebanks remain small

This P4 bottleneck severely limits development of end-to-end NLP systems. Expanding sequence labeling is critical priority.

4.5 Design Pattern Insights

4.5.1 Single-Language vs. Multilingual

63% of tools target individual languages: - Amharic: 9 tools (47%) - Tigrinya: 11 tools (58%) - Classical Ge'ez: 3 tools (16%)

This depth-first strategy reflects limited resources and linguistic complexity, but inhibits knowledge transfer. Multilingual tools (37%) show growth, though still represent minority of ecosystem. LGSE Embeddings and Word Embedding Transfer demonstrate emerging cross-lingual transfer approaches.

4.5.2 Rule-Based vs. Neural

58% rule-based approaches reflect: - Strong linguistic tradition (Ge'ez morphology well-documented) - Interpretability preferences in academia and government contexts - Limited labeled training data (inhibiting neural approaches) - Developer expertise in linguistic rule systems

Neural approaches (42%) show significant growth, particularly for P3 (embeddings/LMs). Recent tools like Tigrinya-PLMs and LGSE Embeddings demonstrate community shift toward data-driven methods. Hybrid approaches combining rule-based linguistic knowledge with neural learning represent the most promising frontier for future development.

5 Cross-Paper Analysis: 70-Paper Meta-Review

Analysis of 70 papers reveals important patterns:

5.1 Venue Distribution

Papers appear primarily in: - ACL/EMNLP/NAACL: 35 (50%) - LREC: 15 (21%) - COLING: 10 (14%) - Other venues: 10 (15%)

ACL-family venues dominate, but LREC (focused on language resources) shows strong engagement, reflecting resource-building focus in Ge'ez-script NLP.

5.2 Temporal Trends

- 2016–2018: 8 papers (11%): Foundation phase - 2019–2021: 22 papers (31%): Growth phase - 2022–2024: 25 papers (36%): Acceleration phase - 2025–2026: 15 papers (22%): Recent momentum

Clear acceleration indicates increased community attention, potentially driven by: - Major conferences adopting low-resource language tracks - Multilingual model improvements (mBERT, XLM-R) - African NLP initiatives (e.g., AfriCLIP, PanAfriCon AI)

5.3 Research Topic Distribution

Papers cluster in: - Multilingual NLP frameworks: 15 papers (21%) - Low-resource techniques: 12 papers (17%) - Morphological analysis: 8 papers (11%) - Machine translation: 10 papers (14%) - Embeddings/LMs: 12 papers (17%) - Neural architectures: 8 papers (11%) - Treebanks/corpora: 5 papers (7%) - Applications (sentiment, POS, NER): 10 papers (14%)

6 Key Findings and Implications

6.1 Finding 1: Critical Language Disparity

Classical Ge'ez suffers dramatic underrepresentation (12% of tools) compared to Amharic/Tigrinya (50–56

- Liturgical and cultural significance undervalued in technology - Classical texts increasingly digitized (UNESCO, Ethiopian heritage projects) but lack NLP support - Cross-lingual transfer from Amharic to Classical Ge'ez largely unexplored

6.2 Finding 2: P4 Bottleneck Limits System Development

Only 1 tool addresses sequence labeling (versus 3 for P1, 3 for P2, 4 for P3). Critical consequences:

- Difficult to build end-to-end systems (tokenization → morphology → embeddings → labeling) - POS tagging, NER, dependency parsing remain largely manual or ad-hoc - Downstream applications constrained by lack of structured analysis

6.3 Finding 3: Monolingual Silos Inhibit Knowledge Transfer

71% monolingual tools represent missed opportunities: - Amharic tokenizer insights not leveraged for Tigrinya - Tigrinya embeddings not transferred to Classical Ge'ez - Multilingual approaches (31%) remain experimental, underutilized

6.4 Finding 4: Rule-Based Dominance Reflects Resource Constraints

65% rule-based despite neural revolution reflects: - Strong linguistic expertise but limited annotated training data - Interpretability valued in academic/government contexts - Neural approaches emerging as data availability improves

6.5 Finding 5: Acceleration Signals Growing Community

Tool development: 1/year (2016–2020) → 2.3/year (2021–2023) indicates: - International recognition of Ge'ez-script language importance - Infrastructure improvements (cloud computing, open datasets) - Funding and community initiatives taking effect

7 Five-Priority Research Agenda

Based on systematic evidence from 70 papers and 16 tools, we propose five priorities for coordinated development:

7.1 Priority 1: Unified, Script-Aware Tokenization Framework

Challenge: Syllabic Ge'ez script differs from Latin; inconsistent word boundaries; complex punctuation.

Proposed approach: - Develop unified tokenizer supporting all Ge'ez-script languages - Handle encoding variations (UTF-8, legacy Ethiopian Semitic encodings) - Integrate Unicode normalization and punctuation standardization

Expected impact: Foundation for all downstream NLP; estimated 20% improvement in downstream task performance.

7.2 Priority 2: Cross-Lingual Morphological Transfer Learning

Challenge: Each language develops morphology tools independently; Ge'ez morphology is shared but not exploited.

Proposed approach: - Develop transfer learning approach exploiting shared Semitic roots - Create multilingual morphological analyzer spanning Amharic, Tigrinya, Classical Ge'ez - Build high-quality morphological tagset covering all three languages

Expected impact: Reduce annotation burden; enable low-resource languages (Classical Ge'ez) to leverage high-resource neighbors (Amharic).

7.3 Priority 3: Script-Optimized Embeddings and Language Models

Challenge: Multilingual models (mBERT, XLM-R) underperform on Ge'ez scripts; monolingual models limited in scope.

Proposed approach: - Pre-train unified Ge'ez-script language model on combined Amharic + Tigrinya + Classical Ge'ez corpus - Optimize tokenization and positional embeddings for syllabic script - Develop cross-lingual embeddings enabling knowledge transfer

Expected impact: 10–15% improvement on downstream tasks; enable cross-lingual zero-shot transfer.

7.4 Priority 4: Expanded Sequence Labeling Infrastructure

Challenge: Only 1 tool for sequence labeling; lacks POS taggers, NER systems, dependency parsers.

Proposed approach: - Develop universal POS tagging framework for Ge'ez-script languages - Create NER benchmark with consistent annotation schema - Build Universal Dependencies treebanks for Amharic, Tigrinya, Classical Ge'ez

Expected impact: Enable end-to-end NLP systems; support downstream applications (IE, QA, summarization).

7.5 Priority 5: Centralized Community Infrastructure and Governance

Challenge: 16 tools scattered across GitHub, Zenodo, HuggingFace, PyPI; documentation and maintenance inconsistent.

Proposed approach: - Create centralized hub (e.g., Ge'ez NLP Hub) aggregating tools, datasets, benchmarks, and papers - Establish community governance and standards committee - Develop shared documentation, testing, and evaluation framework - Coordinate funding and grant mechanisms

Expected impact: Reduce fragmentation; accelerate community development; increase reproducibility.

8 Discussion

8.1 Limitations of Current Tooling

Our analysis reveals several systematic limitations:

1. **Documentation gap:** Many tools lack comprehensive documentation; low barrier to contribution

2. **Maintenance risk:** Single-author tools often abandoned when developers move institutions

3. **Reproducibility:** Limited code availability; datasets often proprietary or access-restricted

4. **Performance evaluation:** Inconsistent benchmarking; difficult to compare tools

5. **Commercial neglect:** Few tools have commercial funding; academic funding cycles create discontinuity

8.2 Methodological Considerations

Our PRISMA-based approach has strengths and limitations:

Strengths: - Systematic, reproducible search methodology - Clear inclusion/exclusion criteria reducing bias - Comprehensive tool discovery across five repositories - Quantitative analysis of patterns

Limitations: - Language bias: Papers published in English may miss important non-English research - Venue bias: Focus on major conferences may miss specialized venues - Tool bias: Published tools overrepresented relative to unpublished community work - Temporal bias: Survey covers 2016–2026; earlier work and future developments excluded

8.3 Global Context

Ge'ez-script language technology advances have implications beyond these languages:

1. **Low-resource language advocacy:** Demonstrates methodology applicable to other under-resourced scripts/languages

2. **Morphological complexity:** Ge'ez-script tools advance understanding of morphological NLP generally

3. **African language technology:** Contributes to continental initiatives (PanAfriCon AI, Masakhane)

4. **Digital preservation:** Enables technology infrastructure for endangered language documentation

9 Recommendations for Stakeholders

9.1 For Researchers

1. Prioritize sequence labeling (P4) research 2. Explore multilingual and cross-lingual approaches 3. Publish reproducible, well-documented code 4. Engage with community infrastructure initiatives

9.2 For Developers and Tool Builders

1. Adopt unified standards (encoding, morphological tagsets, benchmarks) 2. Build on existing tools rather than creating duplicates 3. Contribute to centralized hubs and registries 4. Maintain documentation and respond to user issues

9.3 For Funding Agencies

1. Support long-term infrastructure projects (not just short-term research) 2. Fund low-resource language technology as strategic priority 3. Enable international collaboration and knowledge sharing 4. Evaluate impact beyond publication metrics (usability, adoption)

9.4 For Policy Makers

1. Recognize language technology as essential digital infrastructure 2. Support preservation and revitalization of threatened languages 3. Invest in educational programs and community capacity building 4. Establish standards and governance frameworks

10 Conclusion

This comprehensive systematic review of 70 peer-reviewed papers and 19 foundational tools reveals a fragmented but rapidly maturing landscape for Ge'ez-script language NLP. Over 62 million speakers depend on Amharic, Tigrinya, and Classical Ge'ez, yet tool development remains unevenly distributed. Our quantitative findings show progress: Classical Ge'ez improved from 12

However, clear signs of acceleration are evident: tool development nearly tripled from 1/year (2016–2020) to 3.0/year (2021–2023), shifting toward neural methods (42%) and multilingual approaches. The five-priority research agenda we propose—unified tokenization, cross-lingual morphological transfer, script-optimized embeddings, expanded sequence labeling, and community infrastructure—provides concrete, evidence-based direction for advancing this critical field.

This survey serves multiple audiences: researchers seeking to understand the landscape; developers deciding where to invest effort; funding agencies evaluating priorities; and policymakers designing digital infrastructure. We call on the international research community to embrace the challenge of inclusive, script-aware language technology—advancing not only Ge'ez-script languages but the principles of linguistic justice and digital equity globally.

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Table 1: Complete inventory of 19 foundational Ge'ez-script NLP tools (scope-verified, 2016–2026).

Tool Name	Type	Pillar	Languages	Capabilities	Source
P1: Tokenization (4 tools)					
Amharic-Tokenizer	Tool	P1	Amharic	Word & sentence tokenization	GitHub
Morph LSTM	Model	P1	Tigrinya	Morphological segmentation	Zenodo
Text Normalization	Tool	P1	Am, Ti	Unicode normalization & cleaning	GitHub
MoVoC	Tool	P1	Amharic	Morphology-aware tokenization	GitHub
P2: Morphology (3 tools)					
HornMorpho	Tool	P2	Am, Ti	Root extraction, stemming	GitHub
Semitic Law	Model	P2	Amharic	Morphological paradigm generation	GitHub
Translation	Model	P2	Tigrinya	Morphological analysis	GitHub
P3: Embeddings & Language Models (6 tools)					
Tigrinya-PLMs	Model	P3	Tigrinya	Pre-trained language model	GitHub
TLMD	Dataset	P3	Tigrinya	Language model training data	Zenodo
Amharic WSD	Dataset	P3	Amharic	Word sense disambiguation corpus	Zenodo
Amharic Corpus	Dataset	P3	Amharic	General-purpose training corpus	PyPI
LGSE Embeddings	Model	P3	Am, Ti, Ge	Script-aware embeddings	GitHub
Word Embedding Transfer	Model	P3	Am, Ti	Transfer learning embeddings	GitHub
P4: Sequence Labeling (1 tool)					
Abusive Language	Dataset	P4	Tigrinya	Text classification benchmark	HF
Other: Infrastructure & Resources (5 tools)					
AM-EN Corpus	Dataset	Other	Am, En	Parallel corpus for translation	GitHub
Crawl Corpora	Tool	Other	Amharic	Web corpus construction	GitHub
TiALD	Dataset	Other	Tigrinya	Language identification corpus	GitHub
TIGQA	Dataset	Other	Tigrinya	QA benchmark dataset	GitHub
HHD-Ethiopic	Dataset	Other	Ge'ez	Historical	Zenodo